

Backend Development with Node.js, Express, and Mongo

CIA-1 Examination Answers

- **Submitted By:**
 - **Reg No.:** 2462332
 - **Name:** Joel Anthony D'silva
 - **Email:** joel.anthony@btech.christuniversity.in
- **Course:** Backend Development with Node.js, Express, and Mongo
- **Instructor Name:** Mr. Thirupathi
- **Institution:** Christ University
- **Date of Submission:** 24/07/2026
- **GitHub Repo:**
<https://github.com/joel-dsilva/L-T-Backend-Node.js-Practical-Programs>
- **Theory Set:** A
- **Practical Set:** 1

Part 1 - Theory Set A

Q1. JS Introduction

JavaScript code can be run directly inside a browser without needing a separate _____ step before execution.

Answer: compilation

Explanation: Since JavaScript is an interpreted language the browser's JavaScript engine reads and runs the code line by line, so there is no separate compilation step needed before the code can execute.

Q2. Variables

Write a program to declare a variable `price` with value 250, then increase it by 50 and print the new value.

Answer:

Code:

```
let price = 250;  
price = price + 50; // or: price += 50;  
  
console.log("New Price:", price);
```

Output:

New Price: 300

Q3. Datatypes

The value `undefined` in JavaScript indicates that a variable has been declared but not _____.

Answer: initialized

Explanation: When a variable is declared using var, let, or const but no value is assigned to it, the value 'undefined' is automatically given until an actual value is assigned.

Q4. Operators

Write a program to check if two numbers are equal using both == and ===, and print both results.

Answer:

The '==' operator checks only the value and while '===' checks both value and datatype.

Code:

```
let a = 5;  
let b = "5";  
  
console.log("Using == :", a == b);  
console.log("Using === :", a === b);
```

Output:

```
Using == : true  
Using === : false
```

Q5. Functions

A function that is defined without a name and assigned to a variable is called a _____ function.

Answer: anonymous

Explanation: An anonymous function has no name of its own.

Part 2 – Practical Set 1

Attempted Tasks: Task 1, Task 2, Task 3, Task 4, Task 5, Task 11, Task 12, Task 13, Task 14, Task 15

Task 1: Welcome Message

Answer:

Code:

```
console.log("Welcome to the Student Record System!");
```

Output:

```
Welcome to the Student Record System!
```

Task 2: Declare and Update Variables

Answer:

Code:

```
let systemName = "Student Record System";  
let totalStudents = 5;  
  
console.log("System Name:", systemName);  
console.log("Total Students:", totalStudents);
```

Output:

```
System Name: Student Record System  
Total Students: 5
```

Task 3: Identify Datatypes

Answer:

Code:

```
let studentMarks = 85;      // Number
let studentName = "Riya";   // String
let isPassed = true;        // Boolean

console.log("studentMarks is a", typeof studentMarks);
console.log("studentName is a", typeof studentName);
console.log("isPassed is a", typeof isPassed);
```

Output:

```
studentMarks is a number
studentName is a string
isPassed is a boolean
```

Task 4: Perform a Calculation

Answer:

Code:

```
let marks1 = 78;
let marks2 = 85;
let marks3 = 92;

let average = (marks1 + marks2 + marks3) / 3;

console.log("Average Marks:", average);
```

Output:

```
Average Marks: 85
```

Task 5: Classify a Record

Answer:

Code:

```
let marks = 55;

if (marks >= 40) {
  console.log("Result: Pass");
} else {
  console.log("Result: Fail");
}
```

Output:

Result: Pass

Task 11: Calculate a Derived Value

Answer:

Code:

```
function getResult(name, marks) {
  if (marks >= 40) {
    return name + " has Passed";
  } else {
    return name + " has Failed";
  }
}

console.log(getResult("Ankit", 76));
console.log(getResult("Meera", 32));
```

Output:

Ankit has Passed
Meera has Failed

Task 12: Arrow Function Version

Answer:

Code:

```
const getResult = (name, marks) => {  
  if (marks >= 40) {  
    return name + " has Passed";  
  } else {  
    return name + " has Failed";  
  }  
};  
  
console.log(getResult("Farhan", 58));  
console.log(getResult("Divya", 21));
```

Output:

Farhan has Passed
Divya has Failed

Task 13: Format Output with Template Literals

Answer:

Code:

```
let student = {  
  name: "Tanya",  
  marks: 82,  
  section: "B"  
};  
  
console.log(`Student ${student.name} from Section ${student.section} scored  
${student.marks} marks.`);
```

Output:

Student Tanya from Section B scored 82 marks.

Task 14: Track a Date

Answer:

Code:

```
let admissionDate = new Date(2023, 5, 15); // Month is 0-indexed, so 5 = June
console.log("Admission Date:", admissionDate.toDateString());
```

Output:

Admission Date: Thu Jun 15 2023

Task 15: Apply a Math Method

Answer:

Code:

```
let rawMarks = 87.6;
let roundedMarks = Math.round(rawMarks);

let marksList = [72, 88, 65, 91, 79];
let highestMarks = Math.max(...marksList);

console.log("Rounded Marks:", roundedMarks);
console.log("Highest Marks in List:", highestMarks);
```

Output:

Rounded Marks: 88
Highest Marks in List: 91

GitHub Repo:

<https://github.com/joel-dsilva/L-T-Backend-Node.js-Practical-Programs>

L&T
5th Sem

CIA - 1
THEORY

Name: Joel Dulu
Class: 5BTCSD
RegNo: 2462332

Q1) JavaScript code can be run directly inside a browser without needing a separate compilation step before execution.

Ans: - Compilation.

- Any browser's JavaScript engine reads and runs the code line by line.
- Hence no separate step to compile is needed

Q2) Write a program to declare a variable 'price' with value 250, then increase it by 50 and print the new value.

Ans: Code -

```
let price = 250;  
price = price + 50;  
console.log("New Price:", price);
```

Output:- New Price: 300

Q3) The value 'undefined' in JavaScript indicates that a variable has been declared but not initialized.

Ans: - initialized

- A value undefined is always given to any variable that is declared using var, let or const. and has no value assigned

Q5) A function that is defined without a name and assigned to a variable is called a anonymous function.

Ans: - anonymous

- An anonymous function has no name of its own (PTO)

Q4 Write a program to check if two numbers are equal only both == and === and print both results.

Ans :- Program: let a = 5
let b = "5"

console.log("Using == : ", a == b);

console.log("Using === : ", a === b);

Output : Using == : true

Using === : false

CIA 1 - PRACTICAL

Name: Joel Anthony
Dsilva

Reg. No: 2462332

Class: BSC
DS

TASK 1: Welcome Message.

Code: `console.log("Welcome to the Student Record System!");`

TASK 2: Declare and Update Variables

Code:

```
let systemName = "Student Record System";  
let totalStudents = 5;  
console.log("System Name:", systemName);  
console.log("Total Students:", totalStudents);
```

Output: System Name: Student Record System
Total Students: 5

TASK 3: Identify Datatypes

Code: `let studentMarks = 85;
let studentName = "Priya";
let isPassed = true;`

```
console.log("studentMarks is a", typeof studentMarks);  
console.log("studentName is a", typeof studentName);  
console.log("isPassed is a", typeof isPassed);
```

Output: studentMarks is a number
studentName is a string
isPassed is a boolean.

TASK 4: Perform a Calculation

Code: `let marks1 = 85;
let marks2 = 78;
let marks3 = 92;
let avg = (marks1 + marks2 + marks3) / 3;
console.log("Avg Marks:", avg);`

Output: Avg Marks: 85

TASK 5: Clarify a Record

Code:
let marks = 55;
if (marks >= 40) {
 console.log("Result : Pass");
} else {
 console.log("Result : Fail");
}

Output: Result: Pass

TASK 11: Calculate a Derived Value

Code:
function getResult (name, marks) {
 if (marks >= 40) {
 return name + " has Passed";
 } else {
 return name + " has Failed";
 }
}
console.log(getResult("Ankit", 76));
console.log(getResult("Meera", 32));

Output: Ankit has Passed
Meera has Failed.

TASK 12: Arrow Function Version

Code:
const getResult = (name, marks) => {
 if (marks >= 40) {
 return name + " has Passed";
 } else {
 return name + " has Failed";
 }
}

console.log(getResult("Farhan", 88));
console.log(getResult("Divya", 21));

Output:
Farhan has passed
Divya has Failed.

TASK 13: Formatted Output with Template Literals

Code: let student = {
 name: "Tanya",
 marks: 82,
 section: "B"

};

console.log(`Student \${student.name} from
 Section \${student.section} scored
 \${student.marks} marks.`);

Output:

Student Tanya from Section B scored
 82 marks.

TASK 14: Track a Date

Code: let admissionDate = new Date(2023, 5, 15);
 console.log("Admission Date:" admissionDate.
 toDateString());

Output: Admission Date: Thu Jun 15 2023

TASK 15: Apply a Math Method

Code: let rawMarks = 87.6;
 let roundedMarks = Math.round(rawMarks);
 let marksList = [72, 88, 65, 91, 79];
 let highestMark = Math.max(...marksList);
 console.log("Rounded Marks:", roundedMarks);
 console.log("Highest Mark in List:", highestMark);

Output: Rounded Marks: 88
 Highest Mark in List: 91